

1/3 Sets in Python

26/8/25

Day 40

Date: / /

Page No.:

$O(n)$

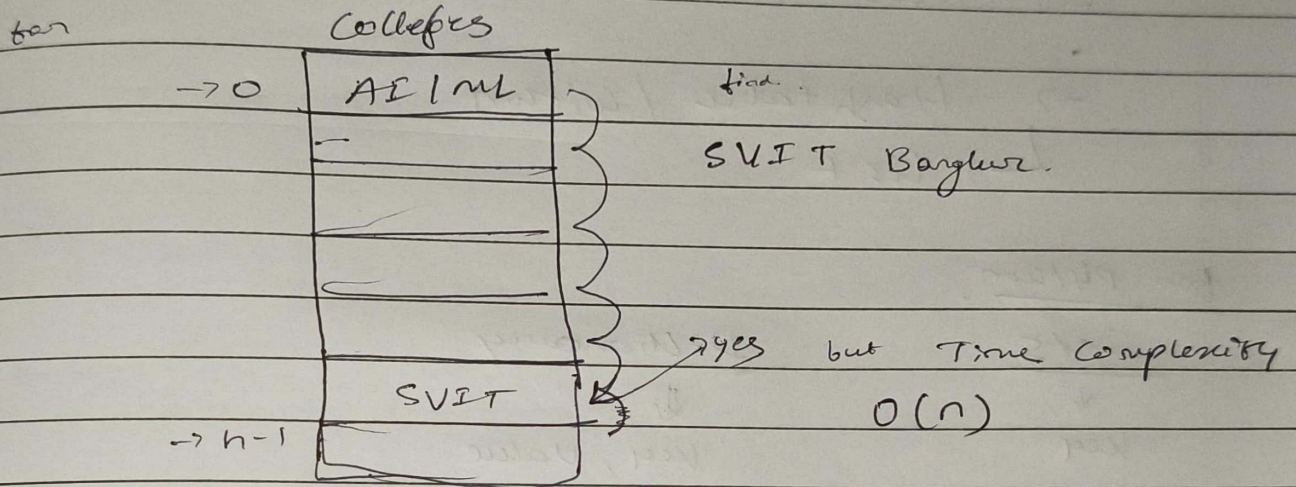
If it starts
X
blon't
no

1	
2	
3	10

numbers [3]

Rear $O(1)$

In List:-

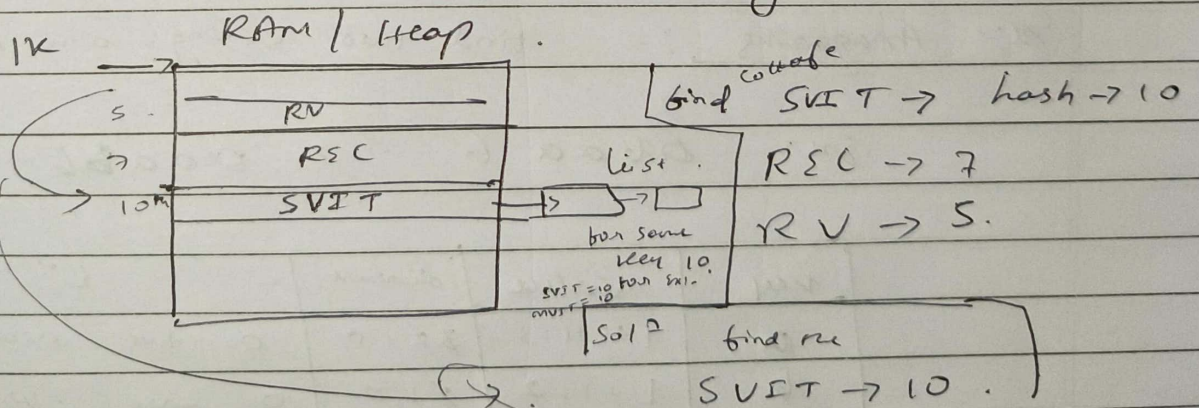


for adjusting this : comes.

Sets / Dictionaries

Value → Hash → alphanumeric.

Storing



∴ Time Complexity is $O(1)$

Some times the
Hash function.

Ex 1 - SVIT $\rightarrow$ 10
MVIT $\rightarrow$ 10

$\rightarrow$ Same hash value key

$\rightarrow$ Hash collision

$\rightarrow$ Hash table / within
 $\hookrightarrow$ list

for python.

Sets

$\downarrow$

key

Dictionary

$\downarrow$

key, value

for other like C. etc.

Hashtable

$\downarrow$

key

Ex: Anagram :- find two strings are anagram

Ex: abaa b $\leftrightarrow$ aabbb

key	value	discount
a	$1+1+1=3$	$3 \neq 10$
b	$1+1=2$	$2 \neq 0$

($\because$ 1 extra a & -1)
0 +ve wrong
b -ve .. wrong.

Using array :-

Hashing

0	0	a
1	0	b
2	0	c
3	0	d

$O(1)$
direct

25

65
↑
ASCII value
Array
 $a - 'a' = 0$

$b - 'a' = 1$

$c - 'a' = 2$

→ Count

→ Add

→ Del

→ Search.

.py Union.py Intersection.py Difference.py Add Set Items.py Remove

6_Data Structure in Python > 07_Sets > Main.py > Fruits

```
1 class Fruits:
2     thisset = {"apple", "banana", "cherry", "pineapple", "mango", "papaya"}
```

I

structure in Python > 07_Sets > Sets_intro.py > Sets

```
class Sets:
```

```
    myset = {"apple", "banana", "cherry"}
    print(myset)
```

```
    # Duplicate values will be ignored:
```

```
    thisset = {"apple", "banana", "cherry", "apple"}
    print(thisset)
```

```
    # True and 1 is considered the same value:
```

```
    thisset2 = {"apple", "banana", "cherry", True, 1, 2}
    print(thisset2)
```

```
    # False and 0 is considered the same value:
```

```
    thisset3 = {"apple", "banana", "cherry", False, True, 0}
    print(thisset3)
```

```
    # Get the number of items in a set:
```

```
    print(len(thisset3))
```

```
    # String, int and boolean data types:
```

```
    set1 = {"apple", "banana", "cherry"}
```

```
    set2 = {1, 5, 7, 9, 3}
```

```
    set3 = {True, False, False}
```

```
    print(set1)
```

```
    print(set2)
```

```
    print(set3)
```

```
    # A set with strings, integers and boolean values:
```

```
    set1 = {"abc", 34, True, 40, "male"}
```

```
    print(set1)
```

```
    # type()
```

```
    myset = {"apple", "banana", "cherry"}
```

```
    print(type(myset))
```

```
    # Using the set() constructor to make a set:
```


6_Data Structure in Python > 07_Sets > Access Items.py > ...

```
1  from Main import Fruits
2
3  # thisset = {"apple", "banana", "cherry"}
4  # or import and write this
5  thisset = Fruits.thisset
6
7  # first printing all names of fruits
8  print("All names of fruits ")
9  print(thisset ,"\n")
10
11 # Loop Sets
12 # Loop through the set, and print the values:
13 for x in thisset:
14     print(x)
15
16
17 # Check if "banana" is present in the set:
18 print("banana" in thisset)
19
20
21 # Check if "banana" is NOT present in the set:
22 print("banana" not in thisset)
```



```
1  from Main import Fruits
2
3  thisset = Fruits.thisset
4
5  # first printing all names of fruits
6  print("All names of fruits ")
7  print(thisset ,"\n")
8
9
10 # Add an item to a set, using the add() method:
11 thisset.add("orange")
12 print(thisset)
13
14 # To add items from another set into the current set,
15 # use the update() method.
16
17 some_more_fruits = {"pineapple", "mango", "papaya"}
18
19 thisset.update(some_more_fruits)
20 print(thisset)
21
22 # Add Any Iterable
23 # The object in the update() method does not have to be a set,
24 # it can be any iterable object (tuples, lists, dictionaries etc.).
25
26 mylist = ["kiwi", "sitaphal"]
27
28 thisset.update(mylist)
29
30 print(thisset)
31
32 # copy
33 set1 = {"a", "b", "c"}
34 set1.copy()
35 print(set1)
```



```
1 # Difference
2 print("the Difference mthd \n")
3
4 set1 = {"apple", "banana", "cherry"}
5 set2 = {"google", "microsoft", "apple"}
6
7 # The difference() method will return a new set
8 # that will contain only the items from the first set
9 # that are not present in the other set.
10 set3 = set1.difference(set2)
11 print(set3) # o/p -> {'cherry', 'banana'}
12
13 # You can use the - operator instead of the difference() method,
14 # and you will get the same result.
15 set3 = set1 - set2
16 print(set3)
17
18 # Note: The - operator only allows you to join sets with sets,
19 # and not with other data types like you can with the difference() method.
20
21 # The difference_update() method will also keep the items
22 # from the first set that are not in the other set,
23 # but it will change the original set instead of returning a new set.
24
25 set1.difference_update(set2)
26 print(set1)
27
28 # Symmetric Differences
29 print("\n the Symmetric Difference mthd \n")
30
31 # The symmetric_difference() method will keep only
32 # the elements that are NOT present in both sets.
33
34 set10 = {"apple", "banana", "cherry"}
35 set20 = {"google", "microsoft", "apple"}
36
```



```
# Intersection intersection.py 7 ...
```

```
print("the Intersection mthd \n")
```

```
# Keep ONLY the duplicates
```

```
set1 = {"apple", "banana", "cherry"}
```

```
set2 = {"google", "microsoft", "apple"}
```

```
set3 = set1.intersection(set2)
```

```
print(set3)
```

```
# You can use the & operator instead of the intersection() method,  
# and you will get the same result.
```

```
# Note The & operator only allows you to join sets with sets,  
# and not with other data types like you can with the intersection() method.
```

```
set4 = set1 & set2
```

```
print(set4)
```

```
# The intersection_update() method will also keep ONLY the duplicates,  
# but it will change the original set instead of returning a new set.
```

```
set1.intersection_update(set2)
```

```
print(set1)
```

```
# The values True and 1 are considered the same value.
```

```
# The same goes for False and 0.
```

```
set1 = {"apple", 1, "banana", 0, "cherry"}
```

```
set2 = {False, "google", 1, "apple", 2, True}
```

```
set4 = set1.intersection(set2)
```

```
print(set4)
```

```
# will change the original set instead of returning a new set.
```

```
set1.intersection_update(set2)
```

```
print(set1)
```


Sets_intro.py Access Items.py Union.py X Intersection.py Difference.py

Data Structure in Python > 07_Sets > Join Sets > Union.py > set1

```
1 set1 = {"a", "b", "c"}
2 set2 = {1, 2, 3}
3 set3 = {"John", "Elena"}
4 set4 = {"apple", "bananas", "cherry"}
5
6 print("\n the Union mthd \n")
7 # Union
8 # The union() method returns a new set with all items from both sets.
9 # The union() and update() methods joins all items from both sets.
10 join_set = set1.union(set2)
11 print(join_set)
12
13 # You can use the | operator instead of the union()
14 # Note: The | operator only allows you to join sets with sets,
15 # and not with other data types like you can with the union() method.
16 join_set = set1 | set2
17 print(join_set)
18
19 # Join Multiple Sets
20 # Join multiple sets with the union() method
21 myset = set1.union(set2, set3, set4)
22 print(myset)
23
24 # When using the | operator, separate the sets with more | operators:
25 myset = set1 | set2 | set3 | set4
26 print(myset)
27
28 # Join a Set and a Tuple
29 x = {"a", "b", "c"}
30 y = (1, 2, 3)
31
32 z = x.union(y)
33 print(z, "\n")
34
35 # Update
36 print("the Update mthd \n")
37 # The update() method inserts the items in set2 into set1:
```


Data Structure in Python > 07_Sets > Remove Set Items.py > ...

```
1  from Main import Fruits
2
3  thisset = Fruits.thisset
4
5  # first printing all names of fruits
6  print("All names of fruits ")
7  print(thisset ,"\n")
8
9
10 # To remove an item in a set, use the remove(), or the discard() method.
11 thisset.remove("banana") # only remove not return
12 thisset.discard("cherry")
13
14 # The return value of the pop() method is the removed item.
15 # Remove a random item by using the pop() method:
16 x = thisset.pop()
17 print("the pop item is:",x)
18
19 print(thisset)
20
21 # The clear() method empties the set:
22 thisset.clear()
23
24 print(thisset)
25
26 # The del keyword will delete the set completely:
27 del thisset
28 print(thisset) # #this will raise an error because the set no longer exists
```